

CHIRAL CHERN–WEIL THEORY

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ABSTRACT. Classical Chern–Weil theory extracts characteristic classes from the curvature of a principal bundle using invariant polynomials. We develop a chiral analogue in the setting of modular Koszul duality for vertex algebras. The ordered bar complex $B^{\text{ord}}(\mathcal{A})$ over the Ran space of a curve X plays the role of a chiral classifying space; the Maurer–Cartan element $\Theta_{\mathcal{A}}^{E_1}$ plays the role of a universal connection, with binary component the collision kernel $K_{\mathcal{A}}^{\text{coll}}(z)$. The homotopy-coinvariant comparison produces $q_n((\Theta_{\mathcal{A}}^{E_1})_{0,n})$, and specified normalized functionals extract scalar shadow data by moment-residue, Sugawara, and composite-sector rules. A represented classical r -matrix appears under $H_{\text{CYBE}}^{\text{rep}}$. As the curve X varies over $\overline{\mathcal{M}}_g$, the fiberwise bar differential has an Arakelov curvature term. For $g \geq 2$, the resulting Theorem D data consist of $\text{Obs}_g^{\text{def}}(\mathcal{A})$, the perfect object $\mathfrak{D}_g^K(\mathcal{A}) = \kappa(\mathcal{A})\lambda_{-1}(\mathbb{E}_g)$, its signed Hodge character, and the stable-graph functional. The stable genus-one component begins with $\text{Obs}_{1,1}^{\text{def}}(\mathcal{A})$. The degree-by-degree scalar coordinates

$$S_n(\mathcal{A}) = \tau_{\mathcal{A},n}^{\text{res}} q_n((\Theta_{\mathcal{A}}^{E_1})_{0,n})$$

form the shadow tower, with $(S_2, S_3, S_4) = (\kappa, \mathfrak{C}, \mathfrak{D})$ on the corresponding scalar channels. The four standard depth types (G/L/C/M) are read from the tuple $(S_3, S_4, \text{termination})$, not from the single downstream discriminant $\Delta = 8\kappa S_4$. For multi-weight algebras, the scalar formula conditionally acquires a cross-channel correction $\partial F_g^{\text{cross}}$, which we compute explicitly for $\mathcal{W}_3$ at genus 2: $\partial F_2(\mathcal{W}_3) = (c+204)/(16c)$.

CONTENTS

1. INTRODUCTION

1.1. The deficiency. The bar complex of a chiral algebra $\mathcal{A}$ on a curve X varies as X moves in the moduli space $\overline{\mathcal{M}}_g$. At genus 0, the bar differential squares to zero. At genus 1, it does not: the fiberwise differential acquires fiber Arakelov curvature, and its pushforward to the base produces the Hodge-class obstruction with coefficient κ .

This curvature carries E_1 operadic indices. For an affine Kac–Moody algebra $V_k(\mathfrak{g})$, the binary collision kernel $K_{k,\mathfrak{g}}^{\text{coll}}(z) = k \Omega_{\text{tr}}/z$ is valued in $\mathfrak{g} \otimes \mathfrak{g}$. Its homotopy coinvariant $q_2(K_{k,\mathfrak{g}}^{\text{coll}})$ remains operator-valued, and the normalized residue trace supplies the scalar coordinate. For the Virasoro algebra at central charge c , the collision kernel $K_{\text{Vir}_c}^{\text{coll}}(z) = (c/2)/z^3 + 2T/z$ contains both vacuum and field-dependent terms; the moment-residue functional gives $\tau_{\text{Vir}_c,2}^{\text{res}} q_2(K_{\text{Vir}_c}^{\text{coll}}) = c/2$.

The question is: what mechanism extracts the scalar shadow data from $K_{\mathcal{A}}^{\text{coll}}$? The answer has a precise classical analogue. In classical Chern–Weil theory, the curvature of a principal G -bundle is $\mathfrak{g}$ -valued, and a G -invariant polynomial kills the internal indices to produce a scalar characteristic class. The chiral

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analogue has two successive maps. The averaging comparison $q_n: \text{End}_{\mathcal{A}}(n) \rightarrow \text{End}_{\mathcal{A}}(n)_{h\Sigma_n}$ produces an equivariant operator class. A specified normalized functional $\tau_{\mathcal{A},n}^{\text{res}}$ then produces its scalar coordinate.

1.2. Results. The main contributions are:

- (1) A systematic parallel between classical Chern–Weil theory and the modular Koszul duality programme (Sections ??–??).
- (2) Explicit computation of q_2 and $\tau_{\mathcal{A},2}^{\text{res}}$ on the collision kernels of the standard families: Heisenberg, Virasoro, affine Kac–Moody (all types $A-G$), principal $\mathcal{W}_N$, and free-field systems (Section ??).
- (3) Identification of the shadow tower $(\kappa, \mathfrak{C}, \mathfrak{Q}, \dots)$ as the sequence of chiral characteristic classes, with the G/L/C/M classification as the classification of chiral classifying spaces by the type of their characteristic ring (Section ??).
- (4) Derivation and explicit computation of the multi-weight cross-channel correction $\delta F_g^{\text{cross}}$, including the first closed-form result $\delta F_2(\mathcal{W}_3) = (c+2\mathfrak{o}_4)/(16c)$ (Section ??).

2. CLASSICAL CHERN–WEIL THEORY

2.1. Connections and curvature. Let G be a Lie group with Lie algebra $\mathfrak{g}$, and let $P \rightarrow M$ be a principal G -bundle over a smooth manifold M . The bundle may be topologically non-trivial, but any single local trivialization sees only $\mathfrak{g}$. To detect the global topology from local data, one needs a device that is defined locally but transforms coherently: a connection.

Definition 2.1. A *connection* on P is a $\mathfrak{g}$ -valued 1-form $\nabla \in \Omega^1(P; \mathfrak{g})$ satisfying:

- (i) $R_g^* \nabla = \text{Ad}(g^{-1}) \nabla$ for all $g \in G$ (equivariance);
- (ii) $\nabla(\xi_P) = \xi$ for all $\xi \in \mathfrak{g}$ (normalization along fibers),

where ξ_P is the fundamental vector field of ξ and R_g is the right G -action on P . In a local trivialization over $U \subset M$, a connection is represented by $A \in \Omega^1(U; \mathfrak{g})$.

Definition 2.2. The *curvature* of a connection ∇ is the $\mathfrak{g}$ -valued 2-form

$$F_{\nabla} = d\nabla + \frac{1}{2}[\nabla, \nabla] \in \Omega^2(M; \text{Ad}(P)).$$

In local coordinates: $F = dA + A \wedge A$.

The curvature measures the failure of parallel transport around infinitesimal loops to be trivial. It is $\mathfrak{g}$ -valued: it carries internal indices from the Lie algebra.

Lemma 2.3 (Bianchi identity). $d_{\nabla} F_{\nabla} = 0$.

Proof. Compute $d_{\nabla} F = d_{\nabla}(d\nabla + \frac{1}{2}[\nabla, \nabla]) = d^2\nabla + \frac{1}{2}[d_{\nabla}\nabla, \nabla] - \frac{1}{2}[\nabla, d_{\nabla}\nabla]$. Using $d^2 = 0$, $d_{\nabla}\nabla = F$, and the Jacobi identity for $\mathfrak{g}$, all terms cancel. $\square$

2.2. The Chern–Weil homomorphism.

Definition 2.4. Let $\text{Sym}^k(\mathfrak{g}^*)^G$ denote the space of G -invariant symmetric k -linear forms on $\mathfrak{g}$. For $P \in \text{Sym}^k(\mathfrak{g}^*)^G$, the *Chern–Weil form* is

$$\text{CW}(P) = P(\underbrace{F_{\nabla}, \dots, F_{\nabla}}_k) \in \Omega^{2k}(M).$$

The form $\text{CW}(P)$ is scalar-valued: the invariant polynomial contracts all $\mathfrak{g}$ -indices. Its construction requires the curvature to carry indices (otherwise there is nothing to contract) and the polynomial to be invariant (otherwise the result would depend on the trivialization).

Theorem 2.5 (Chern–Weil). (i) $\text{CW}(P)$ is closed: $d[\text{CW}(P)] = 0$.

(ii) The cohomology class $[\text{CW}(P)] \in H^{2k}(M; \mathbb{R})$ is independent of the connection ∇ .

(iii) The resulting map $\text{CW}: \text{Sym}^*(\mathfrak{g}^*)^G \rightarrow H^*(M; \mathbb{R})$ is a ring homomorphism.

Proof. (i) The Bianchi identity $d_\nabla F = 0$ and the G -invariance of P imply $d[P(F^k)] = k P(d_\nabla F, F^{k-1}) = 0$.

(ii) Let ∇_0, ∇_1 be connections with curvatures F_0, F_1 . Set $\alpha = \nabla_1 - \nabla_0 \in \Omega^1(M; \text{Ad}(P))$ and interpolate: $\nabla_t = \nabla_0 + t\alpha$, $F_t = F_0 + t d_{\nabla_0} \alpha + t^2 \alpha \wedge \alpha$. Then

$$P(F_1^k) - P(F_0^k) = d \int_0^1 k P(\alpha, F_t^{k-1}) dt.$$

The Chern–Simons transgression $\text{TP} = \int_0^1 k P(\alpha, F_t^{k-1}) dt$ exhibits the difference as exact.

(iii) Multiplicativity of the wedge product. $\square$

Example 2.6 (Abelian: $G = U(1)$). $\mathfrak{g} = \mathbb{R}$, $F = dA$ is already scalar. The unique invariant polynomial (up to scale) is the identity. The first Chern class $c_1 = \frac{1}{2\pi i} [F] \in H^2(M; \mathbb{Z})$ is the sole characteristic class. The classifying space $BU(1) = \mathbb{CP}^\infty$ has $H^*(BU(1)) = \mathbb{Z}[c_1]$.

Example 2.7 ($G = SU(2)$). $\mathfrak{g} = \mathfrak{su}(2)$ is traceless, so $c_1 = 0$. The first non-trivial invariant polynomial is $P(X, Y) = -\frac{1}{8\pi^2} \text{tr}(XY)$, giving the second Chern class $c_2 = -\frac{1}{8\pi^2} [\text{tr}(F \wedge F)]$. On S^4 , $\int_{S^4} c_2 \in \mathbb{Z}$ is the instanton number.

Remark 2.8 (Structure of the invariant ring). For $G = GL_n(\mathbb{C})$, the invariant polynomials are generated by $P_k(X) = \text{tr}(X^k)$, $k = 1, \dots, n$, producing Chern classes $c_1, \dots, c_n$. The number of independent generators of $\text{Sym}^*(\mathfrak{g}^*)^G$ determines the *type* of the classifying space BG .

3. CHIRAL ALGEBRAS, THE BAR COMPLEX, AND THE COLLISION KERNEL

Classical Chern–Weil theory requires three ingredients: a connection, a curvature, and an invariant polynomial. Each has a chiral counterpart, and each requires construction. The connection is the Maurer–Cartan element of the E_1 convolution algebra; its binary twisting component is the collision kernel. The homotopy-coinvariant projection supplies an equivariant operator class, and the normalized residue trace plays the role of an invariant polynomial. A representation satisfying $H_{\text{CYBE}}^{\text{rep}}$ identifies the represented collision kernel with a classical r -matrix.

3.1. Chiral algebras and OPE. Let X be a smooth algebraic curve. A chiral algebra $\mathcal{A}$ on X is a factorization algebra encoding operator product expansions. In local coordinates, the data consists of strong generators $\varphi_1, \dots, \varphi_r$ of conformal weights $h_1, \dots, h_r$, with operator product expansion

$$\varphi_i(z) \varphi_j(w) \sim \sum_{p=1}^{h_i+h_j} \frac{c_{ij}^p(w)}{(z-w)^p}$$

where $c_{ij}^p(w)$ are OPE coefficients, possibly depending on the remaining fields.

3.2. The ordered bar complex.

Definition 3.1. The *ordered bar complex* of $\mathcal{A}$ is

$$B^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$$

where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal of $\mathcal{A}$, s^{-1} is the desuspension ($|s^{-1}v| = |v| - 1$), and T^c is the cofree conilpotent coalgebra with the deconcatenation coproduct. Grading is cohomological ($|d| = +1$).

Warning 3.2. Three distinct errors are common in the bar-complex formula. First, using $\mathcal{A}$ instead of $\bar{\mathcal{A}}$ (the augmentation ideal is essential). Second, using the suspension s instead of desuspension s^{-1} (bar is down). Third, using the tensor algebra T instead of the cofree coalgebra T^c (the deconcatenation coproduct is the coalgebra structure).

On a curve X , the bar complex is a factorization coalgebra on the Ran space $\text{Ran}(X) = \bigsqcup_{n \geq 1} \text{Conf}_n(X)$. The bar differential uses the logarithmic propagator $\eta_{12} = d \log(z_1 - z_2)$ at genus 0.

3.3. The collision kernel from the OPE. The binary bar differential encodes the collision of two generators. The propagator $d \log(z_1 - z_2)$ absorbs one pole from the OPE and gives the collision kernel.

Definition 3.3. The *collision kernel* of $\mathcal{A}$ is the binary component of the E_1 Maurer–Cartan element: $\Theta_{\mathcal{A}}^{E_1} \in \text{MC}(\mathfrak{g}_{\mathcal{A}}^{E_1})$:

$$K_{\mathcal{A}}^{\text{coll}}(z) = (\Theta_{\mathcal{A}}^{E_1})_{0,2} \in \text{End}_{\mathcal{A}}(2) \otimes \mathcal{O}(*\Delta).$$

It is obtained from the OPE data by $d \log$ -absorption; its pole order equals $p_{\max}(\text{OPE}) - 1$. Given a representation $\rho: \text{End}_{\mathcal{A}}(2) \rightarrow \text{End}(V^{\otimes 2})$ and $H_{\text{CYBE}}^{\text{rep}}$, the operator $\rho(K_{\mathcal{A}}^{\text{coll}})$ is a classical r -matrix.

We now compute the collision kernel for every standard family.

Computation 3.4 (Heisenberg at level k). Generator J , weight $h = 1$. OPE:

$$J(z)J(w) = \frac{k}{(z-w)^2}.$$

Pole order 2. After $d \log$ -absorption:

$$K_{\text{Heis}_k}^{\text{coll}}(z) = \frac{k}{z}.$$

This is a simple pole. At $k = 0$ the kernel is zero.

Computation 3.5 (Virasoro at central charge c). Generator T , weight $h = 2$. OPE:

$$T(z)T(w) = \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

Pole order 4. After $d \log$ -absorption, only the singular part of the OPE contributes to the collision residue; the regular ∂T term drops (its $d \log$ coefficient vanishes identically). Thus:

$$K_{\text{Vir}_c}^{\text{coll}}(z) = \frac{c/2}{z^3} + \frac{2T}{z}.$$

Cubic pole, *not* quartic. The leading term $(c/2)/z^3$ is a c -number; the subleading term $2T/z$ is field-dependent. The regular term ∂T from the OPE is absent in $K_{\text{Vir}_c}^{\text{coll}}$: it is $d \log$ -absorbed, or equivalently it is a total derivative on the configuration space and contributes 0 to the residue pairing. At central charge $c = 0$, the vacuum cubic term vanishes and the field-dependent piece $2T/z$ remains.

Computation 3.6 (Affine Kac–Moody $V_k(\mathfrak{g})$). Generators J^a , $a = 1, \dots, \dim(\mathfrak{g})$, all weight $h = 1$. Let $(\ , \)$ be the normalized invariant form (long roots squared length 2), $\{t^a\}$ an orthonormal basis. OPE:

$$J^a(z)J^b(w) = \frac{k \delta^{ab}}{(z-w)^2} + \frac{f^{ab}_c J^c(w)}{z-w}.$$

After $d \log$ -absorption:

$$K_{k,\mathfrak{g}}^{\text{coll}}(z) = \frac{k \Omega}{z}, \quad \Omega = \sum_a t^a \otimes t^a \in \mathfrak{g} \otimes \mathfrak{g}.$$

This is the *trace-form convention*. The level k survives d log-absorption. At $k = 0$ the trace-form collision kernel is zero. In the represented *Kazhdan–Lusztig (KZ) convention*, under $H_{\text{CYBE}}^{\text{rep}}$, $r_{k,\mathfrak{g}}^{\text{KZ}}(z) = \Omega_{\text{KZ}} / ((k + h^\vee)z)$ where Ω_{KZ} is the Killing-normalized Casimir; at $k = 0$ this gives $\Omega_{\text{KZ}} / (h^\vee z) \neq 0$ for non-abelian $\mathfrak{g}$ (the Lie bracket persists). The two conventions are *not equal* as rational functions of k : they are related by a level-dependent rescaling $\Omega_{\text{tr}} = (2h^\vee)^{-1} \Omega_{\text{KZ}}$ together with $k \mapsto k(k + h^\vee) / (2h^\vee)$, or equivalently by absorbing the $(k + h^\vee)^{-1}$ factor into the KZ flat connection. The equation $k \Omega_{\text{tr}} = \Omega_{\text{KZ}} / (k + h^\vee)$ should be read as a structural *change of basis* between conventions, not as an identity of rational functions in k : at $k = 0$ the LHS vanishes while the RHS is finite and non-zero. We adopt the trace convention throughout this note; every affine collision kernel in this section carries its level factor explicitly.

Computation 3.7 (Principal $\mathcal{W}_N$ algebra). Generators $W^{(s)}$ at weights $s = 2, 3, \dots, N$ ($N - 1$ generators). The self-OPE of the spin- s primary in the Fateev–Lukyanov normalization has leading singularity

$$W^{(s)}(z) W^{(s)}(w) = \frac{c/s}{(z - w)^{2s}} + \dots$$

After d log-absorption, the vacuum leading term is

$$K_{ss}^{\text{vac}}(z) = \frac{c/s}{z^{2s-1}} + \dots$$

The cross-OPE $W^{(s_1)}(z) W^{(s_2)}(w)$ for $s_1 \neq s_2$ has leading coefficient that is field-dependent, not a c -number.

Computation 3.8 (Fermionic bc system at weight λ). Generators: b (weight λ) and c (weight $1 - \lambda$), both fermionic. Central charge $c_{bc}(\lambda) = 1 - 3(2\lambda - 1)^2$. The singular OPEs:

$$b(z) c(w) \sim \frac{1}{z - w}, \quad b(z) b(w) \sim 0, \quad c(z) c(w) \sim 0.$$

The only singular OPE has a simple pole (pole order 1). After d log-absorption the cross-contraction has pole order 0. It does not contribute to the scalar collision residue:

$$K_{bc}^{\text{coll}}(z) = 0.$$

The self-OPEs $b(z)b(w)$ and $c(z)c(w)$ are regular (fermionic statistics forbid self-contraction at generic λ), so $K_{bb}^{\text{coll}} = K_{cc}^{\text{coll}} = 0$.

At $\lambda = 2$: the reparametrization ghost system of the bosonic string, with $c_{bc}(2) = -26$. At $\lambda = 1/2$: a single Dirac fermion, $c_{bc}(1/2) = 1$.

Computation 3.9 (Bosonic $\beta\gamma$ system at weight λ). Generators: β (weight λ) and γ (weight $1 - \lambda$), both bosonic. Central charge $c_{\beta\gamma}(\lambda) = 2(6\lambda^2 - 6\lambda + 1) = -c_{bc}(\lambda)$. OPE:

$$\beta(z) \gamma(w) \sim \frac{1}{z - w}.$$

After d log-absorption the cross-contraction has pole order 0, so $K_{\beta\gamma}^{\text{coll}}(z) = 0$ in the collision-residue shadow. The modular characteristic comes from the stress tensor and composite sector.

Complementarity check: $c_{\beta\gamma}(\lambda) + c_{bc}(\lambda) = 0$ for all λ . At $\lambda = 1$: $c_{\beta\gamma}(1) = 2$, $c_{bc}(1) = -2$, sum = 0. At $\lambda = 2$: $c_{\beta\gamma}(2) = 26$, $c_{bc}(2) = -26$, sum = 0 (bosonic string ghost cancellation).

4. THE AVERAGING MAP AND THE MODULAR CHARACTERISTIC

4.1. Definition of the averaging map.

Definition 4.1 (Averaging comparison). An *averaging datum* is an R -twisted descent datum from the ordered (E_1) convolution Lie algebra to the modular symmetric convolution Lie algebra. With such a datum fixed, the arity- n comparison is the quotient to Σ_n -coinvariants:

$$\text{av}: \mathfrak{g}_{\mathcal{A}}^{E_1} \longrightarrow \mathfrak{g}_{\mathcal{A}}^{\text{mod}}.$$

For finite-dimensional aritywise calculations over characteristic zero, the Reynolds operator

$$\text{Reyn}(\phi)(g, n) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \sigma \cdot \phi(g, n)$$

identifies invariants with coinvariants after a choice of splitting; it is not a canonical dg Lie section.

Theorem 4.2 (Averaging on the descent locus). *On the descent locus, the comparison map satisfies:*

- (i) *av is a dg Lie morphism.*
- (ii) *$\text{av}(\Theta_{\mathcal{A}}^{E_1}) = \Theta_{\mathcal{A}}$: the ordered MC element maps to the symmetric MC element.*
- (iii) *In arity n , the image is the equivariant operator class $q_n((\Theta_{\mathcal{A}}^{E_1})_{g,n})$; its scalar coordinate is defined after choosing $\tau_{\mathcal{A},n}^{\text{res}}$.*

At the level of MC elements, the ordered MC element has degree-wise components $(K_{\mathcal{A}}^{\text{coll}}, (\Theta_{\mathcal{A}}^{E_1})_{0,3}, (\Theta_{\mathcal{A}}^{E_1})_{0,4}, \dots)$. The Maurer–Cartan equation gives their operadic boundary relations. Under $H_{\text{CYBE}}^{\text{rep}}$, the represented binary component satisfies the classical Yang–Baxter equation; the corresponding arity-three comparison carries the represented associator relation.

The modular MC element $\Theta_{\mathcal{A}} = \text{av}(\Theta_{\mathcal{A}}^{E_1})$ has arity- n component $q_n((\Theta_{\mathcal{A}}^{E_1})_{g,n})$ in the homotopy coinvariants.

4.2. The Σ_2 action on a collision kernel. Before computing κ for specific families, we work out what the averaging map does to a generic collision kernel at degree 2, step by step.

At degree 2, the E_1 convolution algebra consists of elements on the ordered configuration space $\text{Conf}_2(X) = \{(z_1, z_2) \in X^2 : z_1 \neq z_2\}$. Let $z = z_1 - z_2$ be the relative coordinate. After choosing a representation on the space V of strong generators, the collision kernel is a section of $\text{End}(V^{\otimes 2})$ over $\text{Conf}_2(X)$. For generators φ_i, φ_j , write

$$K^{\text{coll}}(z) = \sum_{i,j} K_{ij}(z) \cdot e_i \otimes e_j \in \text{End}(V^{\otimes 2}) \otimes \mathbb{C}((z)),$$

where $\{e_i\}$ is a basis for V and each $K_{ij}(z)$ is a Laurent series in z (possibly with field-dependent coefficients).

The group Σ_2 acts on $\text{Conf}_2(X)$ by swapping: $(z_1, z_2) \mapsto (z_2, z_1)$, equivalently $z \mapsto -z$. Simultaneously, Σ_2 acts on $\text{End}(V)^{\otimes 2}$ by swapping tensor factors: $e_i \otimes e_j \mapsto e_j \otimes e_i$. The combined action on the collision kernel is:

$$(\sigma \cdot K^{\text{coll}})(z) = \sum_{i,j} K_{ij}(-z) \cdot e_j \otimes e_i.$$

Under the characteristic-zero Reynolds identification, the Σ_2 -coinvariant class has representative

$$q_2(K^{\text{coll}})(z) = \frac{1}{2} (K^{\text{coll}}(z) + \sigma \cdot K^{\text{coll}}(z)) = \frac{1}{2} \sum_{i,j} (K_{ij}(z) + K_{ji}(-z)) \cdot e_i \otimes e_j.$$

4.3. From function to scalar: the cyclic trace. The class $(??)$ is an $\text{End}(V^{\otimes 2})$ -valued function of z . Residue extraction and the chosen cyclic trace combine into $\tau_{\mathcal{A},2}^{\text{res}}$.

Residue extraction. The modular convolution algebra at degree 2 pairs with the propagator $\eta_{12} = d \log(z_1 - z_2)$, which extracts the residue at $z = 0$. The relevant coefficient is

$$\text{Res}_{z=0} [q_2(K^{\text{coll}})(z)] = \frac{1}{2} \sum_{i,j} (\text{Res}_{z=0} K_{ij}(z) + \text{Res}_{z=0} K_{ji}(-z)) \cdot e_i \otimes e_j.$$

For an even-order pole $K_{ij}(z) = c/z^{2m}$: $\text{Res}_{z=0}(c/z^{2m}) = 0$ and $\text{Res}_{z=0}(c/(-z)^{2m}) = 0$. For an odd-order pole $K_{ij}(z) = c/z^{2m+1}$: $\text{Res}_{z=0}(c/z^{2m+1}) = 0$ unless $m = 0$ (simple pole), and $\text{Res}_{z=0}(c/(-z)^{2m+1}) = (-1)^{2m+1} \text{Res}_{z=0}(c/z^{2m+1}) = -\text{Res}_{z=0}(c/z^{2m+1})$.

For a simple-pole collision kernel $K_{ij}(z) = c_{ij}/z$:

$$\text{Res}_{z=0} c_{ij}/z + \text{Res}_{z=0} c_{ji}/(-z) = c_{ij} - c_{ji}.$$

This is the *antisymmetric* part of the coefficient matrix. For a symmetric coefficient ($c_{ij} = c_{ji}$, as for $k \delta^{ab}$ in the Heisenberg/KM double pole), the two terms cancel!

The resolution: for higher-weight generators, the relevant coefficient is not the simple residue but the *moment residue*, as we now explain.

The moment and the cyclic trace. The cyclic trace on $\text{End}(V^{\otimes 2})$ incorporates the conformal weights of the generators. For a generator φ_i of weight h_i , the OPE pairing carries a weight factor z^{2h_i-2} from the conformal Ward identity: the two-point function $\langle \varphi_i(z) \varphi_i(0) \rangle \propto z^{-2h_i}$ contributes a volume factor z^{2h_i} , and the propagator $d \log z$ contributes z^{-2} , giving a net factor z^{2h_i-2} .

The cyclic trace therefore evaluates the $(2h_i-2)$ -th *moment* of the residue:

$$\tau_{\mathcal{A},2}^{\text{res}} q_2(K_{\mathcal{A}}^{\text{coll}}) = \sum_i \text{Res}_{z=0} [z^{2h_i-2} \cdot K_{ii}^{\text{vac}}(z)] + (\text{Sugawara shift}).$$

For a single generator of weight h with vacuum collision kernel $K^{\text{vac}}(z) = c_h/z^{2h-1} + \dots$ (leading vacuum pole), the moment residue is:

$$\text{Res}_{z=0} [z^{2h-2} \cdot c_h/z^{2h-1}] = \text{Res}_{z=0} [c_h/z] = c_h.$$

The moment factor z^{2h-2} exactly cancels the excess pole order, reducing any leading pole to a simple pole whose residue is the vacuum OPE coefficient c_h .

Field-dependent terms. The full collision kernel contains field-dependent terms in addition to the vacuum (c -number) part. For Virasoro: $K_{\text{Vir}_c}^{\text{coll}}(z) = (c/2)/z^3 + 2T/z$, where $2T/z$ is field-dependent.

The vacuum projection $K^{\text{vac}}(z) = \langle 0 | K^{\text{coll}}(z) | 0 \rangle$ retains the c -number part. The cyclic trace annihilates field-dependent terms because they have zero vacuum expectation value: $\langle 0 | T | 0 \rangle = 0$. More precisely, the trace on $\text{End}(V^{\otimes 2})$ contracts the two V -slots against the vacuum state, and any term proportional to a non-identity field gives zero after contraction.

For Virasoro at $h = 2$:

$$\text{Res}_{z=0} [z^2 \cdot K_{\text{Vir}_c}^{\text{coll}}(z)] = \text{Res}_{z=0} [z^2 \cdot ((c/2)/z^3 + 2T/z)] = \text{Res}_{z=0} [(c/2)/z + 2Tz] = c/2 + 0.$$

The field-dependent term $2T$ is regular at $z = 0$ after the moment shift, so their residue vanishes. This is a general mechanism: the moment factor z^{2h-2} shifts the vacuum pole to a simple pole and simultaneously makes all field-dependent terms regular.

For families whose scalar shadow is purely diagonal, the extraction rule is

$$\kappa(\mathcal{A}) = \sum_i \kappa_i + \kappa_{\text{sp}},$$

where $\kappa_i = \text{Res}_{z=0} [z^{2h_i-2} \cdot K_{ii}^{\text{vac}}(z)]$ is the per-channel diagonal contribution and κ_{sp} is the Sugawara shift from simple-pole self-contraction (nonzero only for non-abelian same-weight generators). First-order systems and non-principal reductions also contain composite-sector contributions, so (??) is an extraction rule on the diagonal standard families, not a universal trace-residue theorem.

4.4. Explicit trace: $\mathfrak{sl}_2$ at level k . We carry out the full computation for $\mathfrak{g} = \mathfrak{sl}_2$, $\dim = 3$, $h^\vee = 2$. The standard basis is

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

with bracket $[e, f] = h$, $[h, e] = 2e$, $[h, f] = -2f$.

The normalized invariant form (long roots squared length 2) is $(x, y) = \frac{1}{2h^\vee} \text{tr}_{\text{ad}}(\text{ad}(x) \text{ad}(y)) = \frac{1}{4} \text{tr}_{\text{ad}}(\text{ad}(x) \text{ad}(y))$. Compute: $(e, f) = 1$, $(h, h) = 2$, $(e, e) = (f, f) = 0$. The form is indefinite; no real orthonormal basis exists.

The Casimir is $\Omega = \sum_a t^a \otimes t_a$ where $\{t^a\}$ and $\{t_a\}$ are dual bases for $(,)$. With $t^1 = e$, $t^2 = f$, $t^3 = h/2$ and $t_1 = f$, $t_2 = e$, $t_3 = h/2$ (so that $(t^a, t_b) = \delta_b^a$):

$$\Omega = e \otimes f + f \otimes e + \frac{1}{2} h \otimes h.$$

Check: $\text{tr}(\Omega) = (e, f) + (f, e) + \frac{1}{2}(h, h) = 1 + 1 + 1 = 3 = \dim(\mathfrak{sl}_2)$.

The collision kernel is $K_{k, \mathfrak{sl}_2}^{\text{coll}}(z) = k \Omega_{\text{tr}}/z$.

Double-pole channel. Each generator J^a has weight $h = 1$; the moment factor is $z^0 = 1$. The double-pole OPE $J^a(z) J_a(w) \sim k(t^a, t_a)/(z-w)^2 = k/(z-w)^2$ gives collision-kernel coefficient k/z per dual pair. Three dual pairs, total double-pole trace:

$$\text{tr}(d_{\text{fib}}^2|_{\text{dp}}) = 3k \cdot \omega_1.$$

Normalizing by $2h^\vee = 4$: $\kappa_{\text{dp}} = 3k/(2h^\vee) = 3k/4$.

Simple-pole channel. Use the normalized invariant form with dual bases $\{x_i\}$ and $\{x^i\}$, so

$$\Omega = \sum_i x_i \otimes x^i = e \otimes f + f \otimes e + \frac{1}{2} h \otimes h.$$

The adjoint Casimir identity is

$$\sum_i [x_i, [x^i, y]] = 2h^\vee y.$$

For $\mathfrak{sl}_2$, $h^\vee = 2$; with $x_i = e, f, h$ and $x^i = f, e, h/2$, the identity gives $\sum_i [x_i, [x^i, e]] = 4e$. This is the simple-pole self-contraction responsible for the Sugawara shift.

The Sugawara trace:

$$\kappa_{\text{sp}} = \frac{\dim(\mathfrak{g})}{2} = \frac{3}{2}.$$

Total.

$$\kappa(V_k(\mathfrak{sl}_2)) = \kappa_{\text{dp}} + \kappa_{\text{sp}} = \frac{3k}{4} + \frac{3}{2} = \frac{3(k+2)}{4} = \frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee}. \quad \checkmark$$

At $k = 1$: $\kappa = 9/4$. At $k = -2$ (critical): $\kappa = 0$. At $k = 0$: $\kappa = 3/2 = \dim(\mathfrak{sl}_2)/2$ (the Sugawara contribution from the three generators).

4.5. Explicit trace: $\mathfrak{sl}_3$ at level k . For $\mathfrak{g} = \mathfrak{sl}_3$; $\dim = 8$, $h^\vee = 3$. The Lie algebra has the Chevalley basis $\{e_1, e_2, e_3, f_1, f_2, f_3, h_1, h_2\}$ (three positive roots, three negative roots, two Cartan generators), giving eight generators J^a for $a = 1, \dots, 8$.

The collision kernel is $K_{k, \mathfrak{sl}_3}^{\text{coll}}(z) = k \Omega_{\text{tr}}/z$ with $\Omega_{\text{tr}} = \sum_{a=1}^8 t^a \otimes t_a$.

Double-pole channel. Each of the 8 generators has weight 1, moment factor $z^0 = 1$, and per-channel residue $\kappa_a^{\text{dp}} = k$. Summing and normalizing:

$$\kappa_{\text{dp}} = \frac{8k}{2h^\vee} = \frac{8k}{6} = \frac{4k}{3}.$$

Simple-pole channel. The adjoint Casimir eigenvalue is $C_2^{\text{ad}} = 2h^\vee = 6$. The Sugawara trace:

$$\kappa_{\text{sp}} = \frac{\dim(\mathfrak{g})}{2} = \frac{8}{2} = 4.$$

Total.

$$\kappa(V_k(\mathfrak{sl}_3)) = \frac{4k}{3} + 4 = \frac{4k + 12}{3} = \frac{4(k + 3)}{3} = \frac{\dim(\mathfrak{g})(k + h^\vee)}{2h^\vee}. \quad \checkmark$$

At $k = 1$: $\kappa = 16/3$. At $k = -3$ (critical): $\kappa = 0$.

4.6. Computation: bc and $\beta\gamma$ systems. The bc and $\beta\gamma$ systems have a distinct-weight structure ($h_b = \lambda \neq 1 - \lambda = h_c$ for $\lambda \neq 1/2$) and their only singular OPE is a cross-OPE.

The bc system. Generators b (weight λ) and c (weight $1 - \lambda$). The cross-contraction has pole order 0 after d log-absorption:

$$\text{Res}_{z=0}[K_{bc}^{\text{coll}}(z)] = 0.$$

The collision residue has *no residue*. The kappa the diagonal collision-kernel extraction gives zero.

How does κ arise? For the bc system, the stress tensor is

$$T_{bc} = (\partial b) c \cdot \lambda + b (\partial c) \cdot (1 - \lambda),$$

a composite field of weight 2. The curvature $d_{\text{fb}}^2 = \kappa \cdot \omega_g$ arises not from the diagonal collision kernel directly but from the Sugawara-type construction: the composite stress tensor generates a Virasoro subalgebra with central charge $c_{bc}(\lambda) = 1 - 3(2\lambda - 1)^2$.

The modular characteristic of the full bc vertex algebra is

$$\kappa(bc_\lambda) = \frac{c_{bc}(\lambda)}{2} = \frac{1 - 3(2\lambda - 1)^2}{2}.$$

This arises from the Virasoro subalgebra generated by T_{bc} : the bar complex of the bc system, as a factorization coalgebra, has fiberwise curvature controlled by the stress tensor, and $\kappa = c/2$ when the curvature is entirely generated by the Virasoro sector.

Boundary checks: $\lambda = 1/2 \Rightarrow c_{bc} = 1, \kappa = 1/2$ (single Dirac fermion). $\lambda = 2 \Rightarrow c_{bc} = -26, \kappa = -13$ (string reparametrization ghost).

The $\beta\gamma$ system. By complementarity $c_{\beta\gamma} + c_{bc} = 0$:

$$\kappa(\beta\gamma_\lambda) = \frac{c_{\beta\gamma}(\lambda)}{2} = \frac{2(6\lambda^2 - 6\lambda + 1)}{2} = 6\lambda^2 - 6\lambda + 1.$$

Checks: $\lambda = 1/2 \Rightarrow \kappa = -1/2$; $\lambda = 2 \Rightarrow \kappa = 13$; $\kappa(bc_\lambda) + \kappa(\beta\gamma_\lambda) = (1 - 3(2\lambda - 1)^2)/2 + (6\lambda^2 - 6\lambda + 1) = 0$ for all λ (Koszul complementarity, conductor $K = 0$).

Remark 4.3. The bc and $\beta\gamma$ systems illustrate a subtlety: when the only singular OPE is a *cross*-OPE (not a self-OPE), the diagonal collision kernel has zero residue. The scalar κ arises from the Sugawara stress tensor of the composite system: the bar complex detects the conformal anomaly through the Virasoro sector, regardless of whether the generators have non-trivial self-OPEs. For free-field systems whose OPE is purely a cross-OPE, $\kappa = c/2$ exactly, where c is the central charge of the stress tensor. This is consistent

with the general formula (??): the ‘‘Sugawara shift’’ subsumes the entire contribution when the diagonal collision kernel is zero.

4.7. Degree-2: diagonal extraction rule.

Proposition 4.4. *For diagonal standard families, the scalar degree-2 contribution is extracted by the weight-dependent moment residue. For a single generator φ_i of weight h_i , the per-channel contribution is*

$$\kappa_i = \text{Res}_{z=0} \left[z^{2h_i-2} \cdot K_{ii}^{\text{vac}}(z) \right].$$

For abelian and Virasoro-type families, $\kappa(\mathcal{A}) = \sum_i \kappa_i$. For non-abelian affine Kac–Moody in the trace-form convention, $\kappa(\mathcal{A}) = \sum_i \kappa_i + \dim(\mathfrak{g})/2$.

4.8. Computation: Heisenberg. $K_{\text{Heis}_k}^{\text{coll}}(z) = k/z, h = 1$.

$$\kappa_J = \text{Res}_{z=0} \left[z^{2 \cdot 1-2} \cdot k/z \right] = \text{Res}_{z=0} [k/z] = k.$$

$$\boxed{\kappa(H_k) = k.}$$

Boundary checks: $k = 0 \Rightarrow \kappa = 0$ (uncurved); $k = 1 \Rightarrow \kappa = 1$.

4.9. Computation: Virasoro. $K_{\text{Vir}_c}^{\text{vac}}(z) = (c/2)/z^3, h = 2$.

$$\kappa_T = \text{Res}_{z=0} \left[z^{2 \cdot 2-2} \cdot (c/2)/z^3 \right] = \text{Res}_{z=0} \left[z^2 \cdot (c/2)/z^3 \right] = \text{Res}_{z=0} [(c/2)/z] = c/2.$$

$$\boxed{\kappa(\text{Vir}_c) = c/2.}$$

Boundary checks: $c = 0 \Rightarrow \kappa = 0$; $c = 13 \Rightarrow \kappa = 13/2$ (self-dual point); $c = 26 \Rightarrow \kappa = 13$.

4.10. Computation: $\mathcal{W}_N$. Generators $\mathcal{W}^{(s)}$ at weights $s = 2, \dots, N$ with per-channel vacuum collision kernel $K_{ss}^{\text{vac}}(z) = (c/s)/z^{2s-1} + \dots$.

For each generator:

$$\kappa_s = \text{Res}_{z=0} \left[z^{2s-2} \cdot (c/s)/z^{2s-1} \right] = \text{Res}_{z=0} [(c/s)/z] = c/s.$$

Sum over generators:

$$\boxed{\kappa(\mathcal{W}_N) = \sum_{s=2}^N \frac{c}{s} = c \cdot (H_N - 1), \quad H_N = \sum_{j=1}^N \frac{1}{j}.$$

Verification at $N = 2$ ($\mathcal{W}_2 = \text{Vir}$): $\kappa = c \cdot (3/2 - 1) = c/2$.

Verification at $N = 3$: $\kappa = c \cdot (1/2 + 1/3) = 5c/6$.

Verification at $N = 4$: $\kappa = c \cdot (1/2 + 1/3 + 1/4) = 13c/12$.

Warning 4.5. $H_{N-1} \neq H_N - 1$. At $N = 2$: $H_1 = 1$ but $H_2 - 1 = 1/2$. The formula uses $H_N - 1$ (subtract the integer 1 from the N -th harmonic number), not H_{N-1} (the harmonic number at $N - 1$).

4.11. Computation: affine Kac–Moody and the Sugawara shift. For $V_k(\mathfrak{g})$ with $\mathfrak{g}$ simple, the trace-form collision kernel is $K_{k,\mathfrak{g}}^{\text{coll}}(z) = k \Omega_{\text{tr}}/z$. The functional $\tau_{k,\mathfrak{g},2}^{\text{res}}$ evaluates its double-pole channel and gives κ_{dp} . A one-loop self-contraction in the simple-pole OPE channel supplies the Sugawara term and completes κ .

The proof proceeds in three steps.

Double-pole channel. The diagonal OPE $J^a(z)J^a(w) \sim k/(z-w)^2$ contributes to the fiberwise curvature d_{fib}^2 on a genus-1 curve:

$$d_{\text{fib}}^2|_{\text{dp}}(J^a \boxtimes J^b) = (-2\pi i) \cdot k (J^a, J^b) \cdot \omega_1,$$

where $\omega_1 = c_1(\mathbb{E})$ is the Hodge class. Taking the trace over the normalized form $\sum_a (J^a, J_a) = \dim(\mathfrak{g})$:

$$\mathrm{tr}(d_{\mathrm{fib}}^2|_{\mathrm{dp}}) = k \cdot \dim(\mathfrak{g}) \cdot \omega_1.$$

Simple-pole channel (one-loop self-contraction). The cross-OPE $J^a(z)J^b(w) \sim f^{ab}_c J^c(w)/(z-w)$ feeds back into a second OPE contraction: J^c is contracted with itself through a second propagator loop, forming a one-loop Feynman diagram.

$$d_{\mathrm{fib}}^2|_{\mathrm{sp}}(J^a \boxtimes J^b) = (-2\pi i) \cdot \frac{1}{2} \sum_{c,d} f^{ac}_d f^{bc}_d \cdot \omega_1.$$

The factor $\frac{1}{2}$ is the symmetry factor ($|\mathrm{Aut}| = 2$ of the one-loop diagram). The contraction $\sum_{c,d} f^{ac}_d f^{bc}_d$ is the adjoint quadratic Casimir eigenvalue:

$$\sum_{c,d} f^{ac}_d f^{bc}_d = C_2^{\mathrm{ad}} \cdot (J^a, J^b) = 2h^\vee \cdot (J^a, J^b).$$

This is the Killing form identity: $K^{ab} = 2h^\vee \delta^{ab}$ in an orthonormal basis. Taking the trace:

$$\mathrm{tr}(d_{\mathrm{fib}}^2|_{\mathrm{sp}}) = \frac{1}{2} \cdot 2h^\vee \cdot \dim(\mathfrak{g}) \cdot \omega_1 = h^\vee \cdot \dim(\mathfrak{g}) \cdot \omega_1.$$

Combining the channels. On a single diagonal component, the two channels give per-component curvature $k + h^\vee$. Expressing the curvature as a multiple of the Casimir $C = \sum_a J^a \otimes J_a$:

$$m_o = \frac{k + h^\vee}{2h^\vee} \cdot C,$$

where $1/(2h^\vee)$ normalizes against C_2^{ad} . Then $\kappa = \mathrm{tr}(m_o)$:

$$\begin{aligned} \kappa &= \frac{k + h^\vee}{2h^\vee} \cdot \mathrm{tr}(C) = \frac{k + h^\vee}{2h^\vee} \cdot \sum_a (J^a, J_a) \\ &= \frac{(k + h^\vee) \cdot \dim(\mathfrak{g})}{2h^\vee}. \end{aligned}$$

The decomposition:

$$\kappa = \underbrace{\frac{k \cdot \dim(\mathfrak{g})}{2h^\vee}}_{\kappa_{\mathrm{dp}} \text{ (tree-level collision-kernel trace)}} + \underbrace{\frac{\dim(\mathfrak{g})}{2}}_{\kappa_{\mathrm{sp}} \text{ (one-loop, Sugawara shift)}}.$$

$$\boxed{\kappa(V_k(\mathfrak{g})) = \frac{\dim(\mathfrak{g}) (k + h^\vee)}{2h^\vee}.$$

	$\mathfrak{g}$	$\dim$	$h^\vee$	κ	$k = 1$	$k = -h^\vee$
	$\mathfrak{sl}_2$	3	2	$3(k+2)/4$	$9/4$	0
	$\mathfrak{sl}_3$	8	3	$4(k+3)/3$	$16/3$	0
Example 4.6 (Consistency checks).	E_6	78	12	$13(k+12)/4$	$169/4$	0
	E_7	133	18	$133(k+18)/36$	$2527/36$	0
	E_8	248	30	$62(k+30)/15$	$1922/15$	0
	G_2	14	4	$7(k+4)/4$	$35/4$	0
	F_4	52	9	$26(k+9)/9$	$260/9$	0

At $k = -h^\vee$ (critical level), $\kappa = 0$: the uncurved fixed point of Feigin–Frenkel. At $k = 0$, $\kappa = \dim(\mathfrak{g})/2$: the Sugawara shift alone.

4.12. When the Sugawara shift vanishes.

Proposition 4.7. *The Sugawara shift κ_{sp} is non-zero if and only if the algebra has multiple generators of the same conformal weight with non-trivial cross-OPE at the simple-pole level. In the scalar-shift-free cases, $\kappa = \tau_{\mathcal{A},2}^{\text{res}} q_2(K_{\mathcal{A}}^{\text{coll}})$.*

Three cases:

- (i) *Single-generator algebras* (Heisenberg, Virasoro): no cross-OPE, $\kappa_{\text{sp}} = 0$.
- (ii) *Distinct-weight multi-generator algebras* ($\mathcal{W}_N$): the c -number contributions come only from diagonal self-OPEs; the cross-OPEs between generators of different weights have field-dependent leading coefficients that do not produce closed loops. $\kappa_{\text{sp}} = 0$.
- (iii) *Same-weight multi-generator algebras* (Kac–Moody): all $\dim(\mathfrak{g})$ generators have weight 1, the structure constants f^{ab}_c create the one-loop self-contraction, and $\kappa_{\text{sp}} = \dim(\mathfrak{g})/2 \neq 0$.

4.13. Summary of collision kernels and κ -values.

Family	$K^{\text{coll}}(z)$ (vacuum leading)	κ	κ_{sp}
H_k	k/z	k	0
Vir_c	$(c/2)/z^3$	$c/2$	0
$V_k(\mathfrak{g})$	$k \Omega_{\text{tr}}/z$	$\dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$	$\dim(\mathfrak{g})/2$
$\mathcal{W}_N$	$(c/s)/z^{2s-1}$ per spin	$c(H_N-1)$	0
bc_λ	0 in collision residue	$(1-3(2\lambda-1)^2)/2$	all from Sug.
$\beta\gamma_\lambda$	0 in collision residue	$6\lambda^2-6\lambda+1$	all from Sug.

5. THE CHIRAL CURVATURE

The averaging comparison sends the ordered collision kernel to the operator class $q_2(K_{\mathcal{A}}^{\text{coll}})$. On abelian and scalar families the moment-residue functional gives $\tau_{\mathcal{A},2}^{\text{res}} q_2(K_{\mathcal{A}}^{\text{coll}}) = \kappa$. For non-abelian affine Kac–Moody algebras the same functional gives κ_{cl} , and the Sugawara term completes the modular characteristic: $\kappa = \kappa_{\text{cl}} + \dim(\mathfrak{g})/2$. The question it forces is: what role does κ play geometrically? The answer: κ is the proportionality constant of a curvature. The bar differential, which squares to zero on $\mathbb{P}^1$, fails to square to zero on higher-genus curves; the failure is proportional to κ . This section derives that formula from the propagator on a genus- g curve.

5.1. Propagators at higher genus. On $\mathbb{P}^1$, the propagator $\eta_{12} = d \log(z_1 - z_2)$ is single-valued and the bar differential satisfies $d_{\text{bar}}^2 = 0$. On a genus- g Riemann surface Σ_g with $g \geq 1$, the function $z_1 - z_2$ is not globally defined. The single-valued (Arakelov-normalized) propagator is

$$\eta_{ij}^{(g)} = \left[\partial_{z_i} \log E(z_i, z_j) + \pi \sum_{\alpha, \beta=1}^g \omega_\alpha(z_i) (\text{Im } \Omega)_{\alpha\beta}^{-1} \text{Im} \left(\int_{z_0}^{z_j} \omega_\beta \right) \right] (dz_i - dz_j),$$

where $E(z, w)$ is the prime form, $\omega_1, \dots, \omega_g$ are the normalized abelian differentials, and Ω is the $g \times g$ period matrix.

At genus 1 ($\Sigma_g = E_\tau$ with period τ):

$$\eta_{12}^{(1)} = \left[\partial_u \log \mathfrak{F}_1(u \mid \tau) + 2\pi i \frac{\text{Im } u}{\text{Im } \tau} \right]_{u=z_1-z_2} (dz_1 - dz_2),$$

where $\mathfrak{F}_1$ is the odd Jacobi theta function. The real-analytic coefficient multiplies the same (1, 0) differential as the theta quotient; its $+2\pi i$ shift under $u \mapsto u + \tau$ cancels the theta quotient's $-2\pi i$ shift, giving a double-periodic Arakelov-normalized propagator.

5.2. The curvature formula.

Theorem 5.1 (Fiber curvature and base obstruction). *The fiberwise bar differential on $B(\mathcal{A})$ over Σ_g satisfies*

$$d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g^{\text{Ar}},$$

where ω_g^{Ar} is the Arakelov $(1, 1)$ -form on the fiber curve. The GRR/Deligne-pairing pushforward, under H_D^K , produces

$$\mathfrak{D}_g^K(\mathcal{A}) = \kappa(\mathcal{A}) \lambda_{-1}(\mathbb{E}_g), \quad \text{ch}_g(\mathfrak{D}_g^K(\mathcal{A})) = (-1)^g \kappa(\mathcal{A}) c_g(\mathbb{E}_g).$$

For $g \geq 2$, the native class $\text{Obs}_g^{\text{def}}(\mathcal{A}) \in H^2(g(\mathcal{A}))$ enters this formula through the comparison package H_D^{tr} .

At genus 1: $\omega_1 = \frac{i}{2\text{Im}(\tau)} dz \wedge d\bar{z}$. At genus $g \geq 2$: ω_g involves the full period matrix $(\text{Im } \Omega)^{-1}$, which is a $g \times g$ matrix (not a scalar).

The curvature factorizes:

$$d_{\text{fib}}^2 = \underbrace{\kappa(\mathcal{A})}_{\text{algebra}} \cdot \underbrace{\omega_g}_{\text{geometry}}.$$

The first factor is intrinsic to the chiral algebra (a quasi-isomorphism invariant, independent of the curve). The second factor is intrinsic to the complex structure of the curve (independent of the algebra). This factorization is a structural feature of the chiral theory with no direct classical Chern–Weil analogue.

5.3. Cardy–Arakelov test for κ . The curvature formula (??) is algebraic: $\kappa(\mathcal{A})$ is extracted by $\tau_{\mathcal{A},2}^{\text{res}} q_2$ from the collision kernel, and ω_g is a geometric form on Σ_g . The Cardy–Arakelov identity exhibits the same κ as the Arakelov-normalised coefficient of the coincident-limit torus two-point function of the stress tensor, thereby providing an independent geometric derivation of κ from $\langle T(z)T(w) \rangle_\tau$.

Conjecture 5.2 (Cardy–Arakelov test for the Virasoro lane). *Let $\mathcal{A}$ be a chirally Koszul algebra whose scalar invariant is the Virasoro lane $\kappa = c/2$, and let $\Sigma_1 = E_\tau$ be an elliptic curve with Arakelov $(1, 1)$ -form $\omega_1 = \frac{i}{2\text{Im}(\tau)} dz \wedge d\bar{z}$ ($\int_{E_\tau} \omega_1 = 1$). Let $\langle T(z)T(w) \rangle_\tau^{\text{reg}}$ denote the torus two-point function regularised by subtracting the $c/2 \cdot (z - w)^{-4}$ holomorphic short-distance singularity and the Sugawara self-energy shift ΔT_{Sug} . Then the modular Virasoro-lane scalar satisfies*

$$\frac{c}{2} = \lim_{w \rightarrow z} \frac{\langle T(z)T(w) \rangle_\tau^{\text{reg}}}{\omega_1(z)} = \text{Res}_{w=z}^{(2)}(\langle T(z)T(w) \rangle_\tau) / \omega_1(z),$$

where $\text{Res}_{w=z}^{(2)}$ denotes the coefficient of $(z - w)^{-2} dz \wedge d\bar{z}$ in the genus-1 Arakelov propagator expansion. The identity is independent of the insertion point z and of the choice of Arakelov trivialisation, and is covariant under $\tau \mapsto (a\tau + b)/(c\tau + d)$.

Remark 5.3 (Proof obligation). The missing step is an Arakelov/Faltings or Bismut–Gillet–Soulé derivation identifying the regularised torus stress-tensor two-point coefficient with the Virasoro lane $c/2$. Stress-tensor two-point functions do not recover the programme’s full level-trace invariant for Heisenberg or affine Kac–Moody without additional current/OPE data.

Remark 5.4 (Per-family verification). Equation (??) is a Virasoro-lane test. For Heisenberg and affine Kac–Moody, the full scalar invariant is extracted from current/OPE trace data and the Sugawara shift, not from the stress-tensor two-point function alone. For principal $\mathcal{W}_N$, the claimed equality with $c(H_N - 1)$ is a separate family computation.

Remark 5.5 (Chiral Chern–Weil content). Equation (??) is the proposed Virasoro-lane chiral Chern–Weil form of Cardy’s identity. On the classical side, a connection’s curvature pairs against an invariant polynomial to yield a Chern class; on the chiral side, the Maurer–Cartan element $\Theta_{\mathcal{A}}^{E_1}$ restricted to Σ_1 pairs against the Arakelov form ω_1 to yield the Virasoro scalar. The full programme invariant still requires the family-specific extraction rule.

5.4. The chiral Chern–Weil map. As $[\Sigma_g]$ varies over $\overline{\mathcal{M}}_g$, the bar complex forms a family of curved cochain complexes. The fibrewise curvature defines the following typed trace data.

Theorem 5.6 (Chiral Chern–Weil;). [Type signature: Open quadrant; modular-trace presentation; levels $4 \rightarrow 5$; package $H_D = (H_D^1, H_D^K, H_D^{\text{tr}}, H_D^{\text{graph}})$.] *For a cyclic chart algebra $\mathcal{A}$ and $g \geq 2$, the native class is $\text{Obs}_g^{\text{def}}(\mathcal{A}) \in H^2(\text{Def}_g(\mathcal{A}))$. Under H_D^K ,*

$$\mathfrak{D}_g^K(\mathcal{A}) = \kappa(\mathcal{A})\lambda_{-1}(\mathbb{E}_g), \quad \text{ch}_g(\mathfrak{D}_g^K(\mathcal{A})) = (-1)^g \kappa(\mathcal{A})\lambda_g$$

with $\lambda_g = c_g(\mathbb{E}_g)$. Under H_D^1 the pointed genus-one trace of the deformation class is $\kappa(\mathcal{A})\lambda_1$; H_D^{tr} supplies its comparison with $\mathfrak{D}_g^K$. Under H_D^{graph} the numerical output is $F_g = \kappa\lambda_g^{\text{FP}} + \partial F_g^{\text{cross}}$.

The chiral Chern–Weil map:

$$\underbrace{\Theta_{\mathcal{A}}^{E_1}}_{\text{MC element}} \xrightarrow{q} \underbrace{\Theta_{\mathcal{A}}^{\text{mod}}}_{\text{homotopy-coinvariant MC element}} \xrightarrow{\text{genus-}g \text{ deformation component}} \underbrace{\text{Obs}_g^{\text{def}} \in H^2(\text{Def}_g)}_{\substack{\text{native deformation class} \\ g \geq 2}} \xrightarrow{H_D^{\text{tr}}} \underbrace{\mathfrak{D}_g^K \in K^0(\overline{\mathcal{M}}_g)}_{\text{perfect object}} \xrightarrow{\text{ch}_g}$$

Remark 5.7 (Comparison with classical Chern–Weil).

Classical	Chiral
Connection ∇	MC element $\Theta_{\mathcal{A}}^{E_1}$
Curvature $F_{\nabla} \in \Omega^2(M; \mathfrak{g})$	$K_{\mathcal{A}}^{\text{coll}}(z) \in \mathfrak{g}_{\mathcal{A}}^{E_1}; d_{\text{fib}}^2 = \kappa \cdot \omega_g$
Inv. polynomial $P \in \text{Sym}^k(\mathfrak{g}^*)^G$	Averaging map $\text{av}: \mathfrak{g}_{\mathcal{A}}^{E_1} \rightarrow \mathfrak{g}_{\mathcal{A}}^{\text{mod}}$
CW(P) = $P(F^k)$	$\tau_{\mathcal{A},2}^{\text{res}} q_2(K_{\mathcal{A}}^{\text{coll}}) = \kappa_{\text{cl}}$
Bianchi $d_{\nabla} F = 0$	MC: $d\Theta + \frac{1}{2}[\Theta, \Theta] = 0$
Char. class $[P(F^k)] \in H^{2k}(M)$	$\text{ch}_g(\mathfrak{D}_g^K) = (-1)^g \kappa \lambda_g$ under H_D^K
Independence of connection	κ is a qi-invariant
$H^*(BG) = \text{Sym}^*(\mathfrak{g}^*)^G$	Shadow tower $(\kappa, \mathfrak{C}, \mathfrak{Q}, \dots)$

Three structural features have no classical analogue: (i) the base curve X itself varies over $\overline{\mathcal{M}}_g$; (ii) the curvature factorizes as algebra $\times$ geometry; (iii) the characteristic classes are graded by genus g , not by degree of an invariant polynomial.

6. THE SHADOW TOWER

The modular characteristic κ is the scalar coordinate $\tau_{\mathcal{A},2}^{\text{res}} q_2(K_{\mathcal{A}}^{\text{coll}})$ in the abelian and scalar lanes. The ordered MC element $\Theta_{\mathcal{A}}^{E_1}$ has components at every degree: the KZ associator r_3 at degree 3, the quartic coherence r_4 at degree 4, and so on. Each carries E_1 operadic indices; each has a Σ_n -coinvariant

shadow. The sequence of these shadows is the *shadow tower*, and its depth classifies chiral algebras into four types.

6.1. The shadow algebra.

Definition 6.1 (Shadow algebra). Let $\mathcal{A}$ be a chiral algebra with modular cyclic deformation complex $\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A})$. The *shadow algebra* is

$$\mathcal{A}^{\text{sh}} := H_{\bullet}(\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A})),$$

equipped with the descended Lie bracket from the convolution bracket on $\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A})$.

Proposition 6.2. *The convolution bracket has degree 0 and maps degree $r_1 \times \text{degree } r_2$ to degree $r_1 + r_2 - 2$ (by gluing two external legs). It descends to a well-defined graded Lie bracket on $\mathcal{A}^{\text{sh}}$, making $(\mathcal{A}^{\text{sh}}, [-, -])$ a bigraded Lie algebra.*

Proof. The bracket $[-, -]$ on $\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A})$ is a chain map: $d[f, g] = [df, g] + (-1)^{|f|}[f, dg]$. Cocycles map to cocycles; the bracket of a cocycle with a coboundary is a coboundary. The descended bracket inherits graded antisymmetry and the Jacobi identity. $\square$

The shadow algebra carries a bigrading by degree $r \geq 2$ and genus $g \geq 0$:

$$\mathcal{A}^{\text{sh}} = \bigoplus_{r \geq 2} \bigoplus_{g \geq 0} \mathcal{A}_{r,g}^{\text{sh}}.$$

6.2. Shadow projections. The individual shadow invariants are the bigraded components:

- $\kappa(\mathcal{A}) = \mathcal{A}_{2,0}^{\text{sh}}$: the *modular characteristic* (degree 2, genus 0).
- $\Delta_{\mathcal{A}} = \mathcal{A}_{2,*}^{\text{sh}}$: the *spectral discriminant* (all genera at degree 2).
- $\mathfrak{C}(\mathcal{A}) = \mathcal{A}_{3,0}^{\text{sh}}$: the *cubic shadow* (degree 3, genus 0).
- $\mathfrak{Q}(\mathcal{A}) = \mathcal{A}_{4,0}^{\text{sh}}$: the *quartic shadow* (degree 4, genus 0), containing S_4 .
- $\text{Sh}_r(\mathcal{A}) = \mathcal{A}_{r,0}^{\text{sh}}$: the degree- r shadow.

Each scalar shadow is obtained by applying the appropriate normalized functional to the corresponding homotopy-coinvariant operator class:

$$(\kappa, \mathfrak{C}, \mathfrak{Q}, \dots) = (\tau_{\mathcal{A},2}^{\text{res}} q_2(K_{\mathcal{A}}^{\text{coll}}), \tau_{\mathcal{A},3}^{\text{res}} q_3((\Theta_{\mathcal{A}}^{E_1})_{0,3}), \tau_{\mathcal{A},4}^{\text{res}} q_4((\Theta_{\mathcal{A}}^{E_1})_{0,4}), \dots).$$

6.3. What the higher-degree objects are. *Degree 2:* the E_1 datum is the collision kernel $K_{\mathcal{A}}^{\text{coll}}$. Its homotopy coinvariant is the operator $q_2(K_{\mathcal{A}}^{\text{coll}})$, and the scalar coordinate is $\tau_{\mathcal{A},2}^{\text{res}} q_2(K_{\mathcal{A}}^{\text{coll}})$. Under $H_{\text{CYBE}}^{\text{rep}}$, its represented image is a classical Yang–Baxter datum.

Degree 3: the E_1 datum is the KZ associator $r_3(z_1, z_2)$, a function on $\text{Conf}_3(X)$ satisfying the pentagon identity. Its image under av is the cubic shadow $\mathfrak{C}$: the irreducible triple-collision invariant.

Degree 4: the E_1 datum is the quartic R -matrix coherence r_4 , satisfying the quartic Stasheff equation. Its image under av contains the quartic contact invariant S_4 .

General degree r : the obstruction class at degree $r + 1$ is

$$o_{r+1}(\mathcal{A}) = (D_{\mathcal{A}} \Theta_{\mathcal{A}}^{\leq r} + \frac{1}{2} [\Theta_{\mathcal{A}}^{\leq r}, \Theta_{\mathcal{A}}^{\leq r}])_{r+1} \in H^2(\mathcal{A}_{r+1,0}^{\text{sh}}).$$

This is a Lie bracket of shadows at degrees $\leq r$: the shadow algebra bracket connects different degree components and drives the MC recursion.

6.4. The discriminant and the G/L/C/M classification.

Definition 6.3. The *discriminant* of a chiral algebra $\mathcal{A}$ is

$$\Delta = 8\kappa \cdot S_4.$$

Theorem 6.4 (Depth classification). *For the standard verified families, the tuple $(S_3, S_4, \text{termination})$ classifies the four coarse depth types:*

Class	Depth	Shadow criterion	Prototype	Δ
G (Gaussian)	2	$S_3 = S_4 = 0$	Heisenberg	0
L (Lie)	3	$S_3 \neq 0, S_4 = 0$	Affine KM	0
C (Contact)	4	$S_3 = 0, S_4 \neq 0, S_{r \geq 5} = 0$	$\beta\gamma$	$\neq 0$ generically
M (Mixed)	∞	non-terminating	Vir, $\mathcal{W}_N$	$\neq 0$

Remark 6.5. The discriminant $\Delta = 8\kappa S_4$ is downstream data. It detects the quartic/contact channel when it is defined; it is not by itself an archetype classifier.

Remark 6.6 (Analogy with classifying spaces). The G/L/C/M classification is the chiral analogue of classifying principal bundles by the type of their structure group. The number of independent shadow invariants (characteristic classes) determines the complexity of the chiral classifying space. Class G (one invariant, κ) is analogous to $BU(1)$ (one class, c_1). Class M (infinitely many invariants) is analogous to $BO(\infty)$ (infinitely many Pontryagin classes).

A structural difference: classical Chern classes are polynomials in a single curvature F (the degree-2 datum); chiral shadow invariants are averages of distinct higher-degree operations r_n . Each shadow captures genuinely new collision structure not determined by lower degrees, unless $\Delta = 0$ and the tower terminates.

Remark 6.7 (Shadow depth and Koszulness). Shadow depth measures how many shadow invariants are non-zero. Koszulness (bar cohomology concentrated in degree 1) is a separate property. All standard families are Koszul; the shadow depths differ. SC-formality ($m_k^{SC} = 0$ for $k \geq 3$) holds if and only if the algebra is class G .

7. MULTI-WEIGHT STRUCTURE AND CROSS-CHANNEL CORRECTIONS

The package H_D^1 gives the pointed genus-one trace $\kappa\lambda_1$. The package H_D^K gives the perfect object and its signed Hodge character at every genus in its scope. The package H_D^{graph} gives the full numerical functional, including the mixed-weight cross-channel summand. The first witness is $\mathcal{W}_3$ at genus 2, where $\partial F_2 = (c+204)/(16c)$. This section derives these results from the graph-sum structure of the genus expansion.

7.1. Per-channel decomposition. Let $\mathcal{A}$ be a modular Koszul chiral algebra with strong generators $\varphi_1, \dots, \varphi_r$ of conformal weights $h_1, \dots, h_r$. Each generator has a per-channel modular characteristic κ_i : the contribution from the (φ_i, φ_i) self-OPE channel. The total is $\kappa(\mathcal{A}) = \sum_{i=1}^r \kappa_i$.

The genus- g free energy is computed by a graph sum over stable graphs Γ of genus g :

$$F_g(\mathcal{A}) = \sum_{\Gamma} |\text{Aut}(\Gamma)|^{-1} \sum_{\sigma: E(\Gamma) \rightarrow \{1, \dots, r\}} A_{\Gamma}(\sigma, \mathcal{A}),$$

where σ assigns a channel to each edge e of Γ .

Definition 7.1. A channel assignment σ is *diagonal* if $\sigma(e) = i$ for all e (all edges carry the same channel). It is *mixed* if at least two edges carry different channels.

7.2. The decomposition theorem.

Theorem 7.2 (Conditional multi-weight genus expansion). *Assume the diagonal genus- g Faber–Pandharipande comparison and the mixed-channel boundary graph expansion of Volume I. With the notation above:*

- (i) Per-channel universality. *The diagonal contribution satisfies*

$$F_g^{\text{diag}}(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}},$$

where $\lambda_g^{\text{FP}} = \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$ is the Faber–Pandharipande number (B_{2g} the Bernoulli number), for all $g \geq 1$ in the diagonal sector.

- (ii) Cross-channel decomposition. *The full free energy decomposes as*

$$F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}(\mathcal{A}),$$

where $\delta F_g^{\text{cross}}$ is a graph sum over mixed-channel boundary graphs of $\overline{\mathcal{M}}_{g,0}$.

- (iii) Genus-1 universality. $\delta F_1^{\text{cross}} = 0$ for all $\mathcal{A}$.
 (iv) Uniform-weight universality. If $h_1 = \dots = h_r$, then $\delta F_g^{\text{cross}} = 0$ at all $g \geq 1$.
 (v) R -matrix scope. No higher-genus mixed-channel R -independence is asserted here. The R -independence statement is restricted to diagonal terms and to genus 1.
 (vi) Genus-2 explicit formula. At genus 2:

$$\delta F_2^{\text{cross}}(\mathcal{A}) = \delta F_2^{\text{sun}}(\mathcal{A}) + \delta F_2^{\theta}(\mathcal{A}) + \delta F_2^{\text{lt}}(\mathcal{A}) + \delta F_2^{\text{bb}}(\mathcal{A}),$$

summing over four contributing boundary strata among the seven stable graphs of $\overline{\mathcal{M}}_{2,0}$.

Proof. (i) On a diagonal assignment $\sigma(e) = i$ for all e , the amplitude factors as $A_\Gamma(\sigma_i, \mathcal{A}) = A_\Gamma(\mathcal{A}_i)$ for the rank-1 algebra $\mathcal{A}_i$ with curvature κ_i . Each rank-1 component has a single generator and hence uniform weight. Summing:

$$F_g^{\text{diag}}(\mathcal{A}) = \sum_{i=1}^r F_g(\mathcal{A}_i) = \sum_{i=1}^r \kappa_i \cdot \lambda_g^{\text{FP}} = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}.$$

- (iii) At genus 1, the unique boundary stratum has a single edge. A single edge admits only r channel assignments, each pure. No mixed-channel assignments exist on a single-edge graph. $\square$

7.3. Why cross-channel corrections arise at genus $g \geq 2$.

Three ingredients are needed for a non-zero $\delta F_g^{\text{cross}}$:

- (a) *Multiple edges on a single graph*, so that different edges can carry different channels. This requires $g \geq 2$, since all genus-1 contributing graphs have a single edge.
- (b) *Distinct weights* on different channels ($h_i \neq h_j$), so that the propagators along edges carry different channel weights in the universal propagator.
- (c) *Non-trivial OPE mixing*: the vertices carry mixed-channel OPE structure constants not controlled by κ alone.

At genus 1, ingredient (a) fails: the single-edge graph cannot carry a mixed assignment. At uniform weight, ingredient (b) fails: all propagators are identical regardless of channel. For free fields, ingredient (c) fails: $m_k = 0$ for $k \geq 3$ at genus 0, so mixed-channel vertices carry no structure constants beyond the two-point function.

7.4. Explicit computation: $\mathcal{W}_3$ at genus 2. $\mathcal{W}_3$ has two generators: T (weight 2) and W (weight 3). Per-channel modular characteristics: $\kappa_T = c/2$, $\kappa_W = c/3$. Total: $\kappa = c/2 + c/3 = 5c/6$.

The four contributing boundary strata at genus 2 give:

$$\partial F_2(\mathcal{W}_3) = \frac{3}{c} + \frac{9}{2c} + \frac{1}{16} + \frac{21}{4c} = \frac{c + 204}{16c}.$$

This is non-zero for $c \neq 0, -204$; for positive central charge it is non-zero.

- At $c = 1$: $\partial F_2 = 205/16$.
- At large c : $\partial F_2 \rightarrow 1/16$ (the leading correction is a constant, independent of c).
- At $c = -204$: $\partial F_2 = 0$ (an isolated zero, not physically relevant).

The four terms arise from distinct boundary mechanisms:

- $3/c$: sunset graph with mixed-weight vertex carrying the T – W OPE coefficient.
- $9/(2c)$: theta graph with three edges at mixed weight assignments $(2, 2, 3)$, $(2, 3, 3)$, and permutations.
- $1/16$: a topological contribution from the difference in Euler characteristics of the weight-2 and weight-3 determinant line bundles over $\overline{\mathcal{M}}_2$, independent of c .
- $21/(4c)$: bridge-loop graph connecting a genus-1 vertex (single-edge loop at weight h_i) to a genus-0 vertex through an edge at weight $h_j \neq h_i$.

Corollary 7.3. *The scalar formula $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$ fails at $g = 2$ for $\mathcal{W}_3$. This resolves the multi-generator universality question in the negative: the scalar formula is not universal for interacting multi-weight algebras.*

7.5. Free-field exactness.

Proposition 7.4 (Free-field exactness). *Let $\mathcal{A}$ be a free-field chiral algebra ($m_k = 0$ for all $k \geq 3$ at genus 0). Then $\partial F_g^{\text{cross}}(\mathcal{A}) = 0$ for all $g \geq 1$. The scalar formula $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ is exact at all genera.*

Three independent mechanisms force this vanishing:

- (1) The genus-0 OPE is purely quadratic, so mixed-channel vertices carry no structure constants beyond the two-point function.
- (2) The Givental R -matrix is trivial for free fields, eliminating R -corrections at genus-0 vertices.
- (3) The propagator factorization identity for free fields collapses the cross-channel graph sum to the diagonal.

8. THE CHIRAL CLASSIFYING SPACE

The preceding sections constructed each ingredient of chiral Chern–Weil theory individually: the collision kernel (the binary connection datum), the homotopy-coinvariant projection, the normalized trace functional (the scalar invariant), the curvature formula, the shadow tower (the characteristic ring), and the multi-weight correction. The forced question: is there a single geometric object that unifies all of these, the way the classifying space BG unifies classical Chern–Weil theory?

8.1. $B(\mathcal{A})$ as classifying object. The ordered bar complex $B^{\text{ord}}(\mathcal{A})$ on $\text{Ran}(X)$, equipped with the deconcatenation coproduct, is a factorization coalgebra. In the deformation-theoretic picture, it is the chiral analogue of the simplicial bar construction $[n] \mapsto G^n$ whose geometric realization is the classifying space BG .

The universal bundle is the twisted tensor product

$$\mathcal{E}^{\text{univ}}(\mathcal{A}) = \mathcal{A} \otimes^{\tau} B(\mathcal{A}),$$

with differential combining the internal differential, the bar differential, and the twisting morphism $\tau: B(\mathcal{A}) \rightarrow \mathcal{A}$. Its acyclicity, $\Omega(B(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ (bar-cobar quasi-isomorphism on the Koszul locus), is the chiral analogue of the contractibility of EG .

8.2. The universal connection. The MC element $\Theta_{\mathcal{A}} \in \text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}})$ is the *universal connection* on the chiral classifying space. The MC equation

$$d\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$$

is the flatness/integrability condition. At genus 0, the universal connection is flat ($d_{\text{fib}}^2 = 0$); the universal bundle is trivializable over $\mathcal{M}_{0,n}$.

At genus $g \geq 2$, the fibrewise curvature is $d_{\text{fib}}^2 = \kappa \omega_g$. The modular trace assigns the native class $\text{Obs}_g^{\text{def}}$, the perfect object $\mathfrak{D}_g^K = \kappa \lambda_{-1}(\mathbb{E}_g)$ under H_D^K , and the numerical graph functional under H_D^{graph} . At genus one, H_D^1 begins with the pointed class $\text{Obs}_{1,1}^{\text{def}}$.

8.3. The total system over moduli. The total system is a family

$$\mathcal{E}_{\text{total}} \longrightarrow B^{\Sigma}(\mathcal{A})|_{\text{Sym}^{\bullet}(C_g/\overline{\mathcal{M}}_g)} \longrightarrow \overline{\mathcal{M}}_g,$$

where $C_g \rightarrow \overline{\mathcal{M}}_g$ is the universal curve, and $B^{\Sigma}(\mathcal{A})$ is the Σ_n -coinvariant (symmetric) bar complex over the symmetric product $\text{Sym}^n(X) = \text{Conf}_n(X)/\Sigma_n$.

The ordered bar $B^{\text{ord}}(\mathcal{A})$ on $\text{Conf}_n(X)$ carries $K_{\mathcal{A}}^{\text{coll}}(z)$ and, on represented comparison surfaces, the corresponding R -operator. The symmetric bar $B^{\Sigma}(\mathcal{A})$ on $\text{Sym}^n(X)$ retains the homotopy-coinvariant operator. The normalized trace gives κ on scalar families and κ_{dp} on non-abelian affine families, where the Sugawara contribution completes κ .

As $[X]$ moves in $\overline{\mathcal{M}}_g$:

- (i) The bar differential depends on X through the prime-form propagator $d \log E(z, w)$ (weight 1).
- (ii) At genus 0: flat ($d_{\text{fib}}^2 = 0$), universal bundle trivializable.
- (iii) At genus $g \geq 1$: curvature $d_{\text{fib}}^2 = \kappa \omega_g$; H_D^K gives $\mathfrak{D}_g^K = \kappa \lambda_{-1}(\mathbb{E}_g)$.
- (iv) At multi-weight, genus $g \geq 2$: H_D^{graph} gives $F_g = \kappa \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}$.

8.4. Crystallization. Classical Chern–Weil theory has three ingredients: a connection ∇ , an invariant polynomial P , and the resulting characteristic class $[P(F^k)]$. Chiral Chern–Weil theory has the same three: the E_1 MC element $\Theta_{\mathcal{A}}^{E_1}$ (whose binary component is $K_{\mathcal{A}}^{\text{coll}}$), the homotopy-coinvariant map q_n , the normalized functional $\tau_{\mathcal{A},n}^{\text{res}}$, and, for $g \geq 2$, the Theorem D objects $\text{Obs}_g^{\text{def}}$, $\mathfrak{D}_g^K$, its signed Hodge character, and F_g . What is new is the variation over moduli: the curvature $d_{\text{fib}}^2 = \kappa \cdot \omega_g$ factorizes as algebra $\times$ geometry, the shadow tower $(\kappa, \mathfrak{C}, \mathfrak{Q}, \dots)$ replaces the finite-rank invariant ring of a compact group with an degree-graded sequence whose depth classifies chiral algebras (G/L/C/M), and the scalar formula fails at genus ≥ 2 for multi-weight algebras with explicit correction $\delta F_2(\mathcal{W}_3) = (c+204)/(16c)$.

9. KOSZUL CONDUCTORS AND SCALAR COMPLEMENTARITY

The chiral Chern–Weil output $\kappa(\mathcal{A})$ also organises Koszul-complementary pairs. Two distinct invariants live on a Koszul-complementary pair:

$$K_c(\mathcal{A}) := c(\mathcal{A}) + c(\mathcal{A}^{\dagger})$$

and the scalar κ -complementarity $K^{\kappa}(\mathcal{A}) := \kappa(\mathcal{A}) + \kappa(\mathcal{A}^{\dagger})$. There is no single ratio converting K_c to K^{κ} : the affine line has $K_c = 2 \dim \mathfrak{g}$ but $K^{\kappa} = 0$, while Virasoro, principal $\mathcal{W}_N$, and Bershadsky–Polyakov use family-specific normalisations. Trinity values: $K_c = 0$ for class G and abelian Kac–Moody,

$K_c = 2 \dim(\mathfrak{g})$ for non-abelian $V_k(\mathfrak{g})$ while $K^\kappa = 0$ on the affine line; $K_c = 26$ for Virasoro (so $K^\kappa = 13$ at self-dual $c = 13$, with $\varrho_{\text{Vir}} = 1/2$); $K_c = 50$ for the Bershadsky–Polyakov algebra BP (the polynomial identity $K_{\text{BP}}(k) \equiv 50$ as a rational function in the level, in the standard Fehily–Kawasetsu–Ridout convention, with the canonical fixed point at $k = -3 = -h^\vee(\mathfrak{sl}_3)$, where conditionally $K_{\text{BP}}^\kappa = 25/3$ and $\varrho_{\text{BP}} = 1/6$); $K = 100$ for principal $\mathcal{W}_3$ (so $K^\kappa = 250/3$ with $\varrho_{\mathcal{W}_3} = 5/6 = H_3 - 1$). The tabulated central charges of the bosonic $\beta\gamma$ and fermionic bc ghost systems provide the universal base: $c^{\beta\gamma}(\lambda) = 2(6\lambda^2 - 6\lambda + 1)$ and $c^{bc}(\lambda) = 1 - 3(2\lambda - 1)^2$, whose sum vanishes identically.

Proposition 9.1 (Verified conductor lanes). *For the verified standard families, central-charge conductors and scalar modular conductors are family-specific quantities. There is no single BRST ghost object canonically attached to every family in the table.*

Proof. For affine $V_k(\mathfrak{g})$, the central-charge conductor and the scalar modular conductor are different quantities:

$$K_c(V_k(\mathfrak{g})) := c(V_k(\mathfrak{g})) + c(V_{-k-2h^\vee}(\mathfrak{g})) = \frac{k \dim \mathfrak{g}}{k + h^\vee} + \frac{(-k - 2h^\vee) \dim \mathfrak{g}}{-k - h^\vee} = 2 \dim \mathfrak{g},$$

whereas

$$K^\kappa(V_k(\mathfrak{g})) := \kappa(V_k(\mathfrak{g})) + \kappa(V_{-k-2h^\vee}(\mathfrak{g})) = \frac{\dim \mathfrak{g}}{2h^\vee} ((k + h^\vee) + (-k - h^\vee)) = 0.$$

The ghost calculation records the central-charge lane; the vanishing scalar lane is the affine complementarity used by the shadow tower. For Vir_c : the (b, c) ghost system has $\lambda = 2$ giving $c^{bc}(2) = -26$; complementarity $\kappa(\text{Vir}_c) + \kappa(\text{Vir}_{26-c}) = c/2 + (26 - c)/2 = 13 = -c_{\text{ghost}}/2$ after the universal factor. For $\mathcal{W}_N$: the reduction from $\mathfrak{sl}_N$ contributes a tower of ghosts at weights $\{2, 3, \dots, N\}$; summation reproduces the tabulated K -value. For BP, the polynomial identity $K_{\text{BP}}(k) \equiv 50$ in the standard Fehily–Kawasetsu–Ridout convention matches the central-charge conductor $K_c^{\text{BP}} = 50$ at every k ; the conditional scalar conductor on the ratio lane is $K_{\text{BP}}^\kappa = 50/6 = 25/3$, and the fixed point $k = -h^\vee(\mathfrak{sl}_3) = -3$ is a pole with principal-value half $25/6$. $\square$

Proposition 9.2 (Scalar complementarity and moment residues at genus zero). *On the verified diagonal standard families, the genus-zero extraction of the scalar invariant uses the moment-residue pairing*

$$\kappa(\mathcal{A}) = \sum_i \text{Res}_{z=0} (z^{2h_i-2} K_{ii}^{\text{vac}}(z) dz) + \kappa_{\text{sp}},$$

with the affine Sugawara shift κ_{sp} added separately. For a Koszul pair, the scalar conductor is then $K^\kappa(\mathcal{A}) = \kappa(\mathcal{A}) + \kappa(\mathcal{A}^\dagger)$.

Remark 9.3 (Epistemic tags and independent-verification conventions). Every collision kernel in this section carries its family parameter explicitly: the Kac–Moody trace-form kernel is $K_{k,\mathfrak{g}}^{\text{coll}}(z) = k \Omega_{\text{tr}}/z$, and the Virasoro kernel $K_{\text{Vir}_c}^{\text{coll}}(z) = (c/2)/z^3 + 2T/z$ has the canonical cubic pole. Every κ value is taken from the landscape census. The scalar conductor values $\{0, 13, 25/3, 250/3\}$ are verified against two disjoint sources under the independent verification protocol: direct computation of $\kappa + \kappa^\dagger$ and ghost tabulation after conversion from K_c to K^κ where a ghost description is available.

Remark 9.4 (Cross-volume bridge). The central and scalar conductor lanes are separate Vol I manifestations of the trace formalism. In Vol III, the scalar normalization reappears on Borcherds–Kac–Moody algebras as $\kappa_{\text{BKM}}(\Phi_N) = c_N(0)/2$ with $N \in \{1, 2, 3, 4, 6\}$ the level of the Fricke-type paramodular form. A numerical equality with a Vol I scalar conductor is not asserted without specifying the Borcherds input Φ_N , the normalization, and the Vol I family being compared.

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